

Scalability & Future Plan

Date: 30 September 2026 **Team ID:** 6ab3e57da238dc999eb3d44d **Project Name:** EduGenie – AI-Powered Educational Assistant

Maximum Marks: 1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority
1	No persistent database or learner accounts	No saved history or long-term progress tracking.	High
2	Single-process lightweight deployment	Not yet optimized for high concurrent traffic.	Medium
3	AI responses depend on external provider availability	Network/API failures can affect live generation.	High
4	No formal authentication / rate limiting	Not ready for unrestricted public deployment.	High

Scalability Plan

Aspect	Current State	Proposed Upgrade
User Load	Local FastAPI application	Deploy behind a production ASGI setup/load balancer and add horizontal scaling.
Data Storage	No persistent application DB	Add PostgreSQL/managed database for accounts, history and progress.
Performance	AI latency varies by provider	Add caching for safe repeat requests, async processing and monitoring.
Security	API key via environment variable; no auth	Add authentication, authorization, rate limiting, secret manager and audit logging.

Future Roadmap

Phase	Planned Enhancement	Target	Expected Impact
Phase 2	User accounts + saved learning history	Next release	Personalized continuity and progress tracking.
Phase 3	PDF/image input + multilingual support	Future release	Broader study-material support and accessibility.
Phase 4	Teacher dashboards, adaptive learning and LMS integration	Longer-term	Institutional use and richer personalization.